PROJECT PLAN AND SOLUTION ARCHITECTURE

Smart Real Estate System

Salesforce CRM and Agentforce Implementation

A governed CRM solution for property inventory, lead qualification, showing coordination, offers, and operational reporting

Prepared by	Rayian Hussain Harboosh
Document status	Version 1.0 - Project documentation
Date	August 27, 2026

Table of Contents

1. Executive Overview 3

2. Scope, Stakeholders, and Requirements 4

3. Solution Architecture 6

4. Data Architecture 8

5. Security and Governance Architecture 10

6. Project Plan and Delivery Schedule 11

7. Risk Management 12

8. Test and Acceptance Strategy 13

9. Deployment, Maintenance, and Troubleshooting 14

10. Conclusion 14

Appendix A. Evidence Inventory 15

Appendix B. Submission Completion Checklist 15

Evidence note: This document is based on the supplied Salesforce screenshots and SFDX project files. Agentforce is described as configured because the supplied Builder screenshot shows Version 2 in Draft status.

1. Executive Overview

The Smart Real Estate System is a Salesforce-based CRM solution that centralizes property inventory, lead qualification, showing management, offers, follow-up tasks, and operational reporting. Agentforce provides conversational assistance by routing user requests through an Agent Router to specialized subagents and invoking governed Salesforce actions backed by Autolaunched Flows. Lightning pages, reports, dashboards, and a custom Lightning Web Component support the daily work of sales agents and managers.

> **Current implementation position:** Core CRM screens, representative records, dashboards, Agentforce routing, subagent instructions, and a deployed LWC bundle are evidenced. Final release acceptance still requires Agentforce activation evidence, security tests, and verification of the LWC on its intended record context.

1.1 Problem Statement

Real estate sales teams lose time manually triaging inbound leads, checking property availability, coordinating showing logistics, and assembling management reports. These disconnected activities delay initial responses, create inconsistent follow-up, increase the risk of scheduling conflicts, and reduce the visibility available to sales managers.

1.2 Business Objectives

- Centralize lead, property, showing, offer, and follow-up information in Salesforce.
- Automate lead preference updates, qualification support, and follow-up task creation to reduce manual data entry.
- Configure Agentforce to support property inquiries, lead qualification, agent availability checks, confirmed showing creation, safe redirection, and human escalation.
- Provide managers with real-time operational visibility through Lightning reports and dashboards.
- Apply Salesforce security controls and validation rules to protect data quality and role-appropriate access.

1.3 Intended Business Value

Business need	System response	Expected value
Slow lead triage	Structured lead preferences, qualification fields, and follow-up automation	More consistent prioritization and agent follow-up
Fragmented property data	Central Property records with status, location, price, and attributes	A single operational source for inventory
Manual scheduling	Availability checking and governed showing creation	Fewer scheduling conflicts and incomplete bookings

Business need	System response	Expected value
Limited management visibility	Central dashboard for inventory, lead funnel, showings, and offers	Faster operational review and decision support

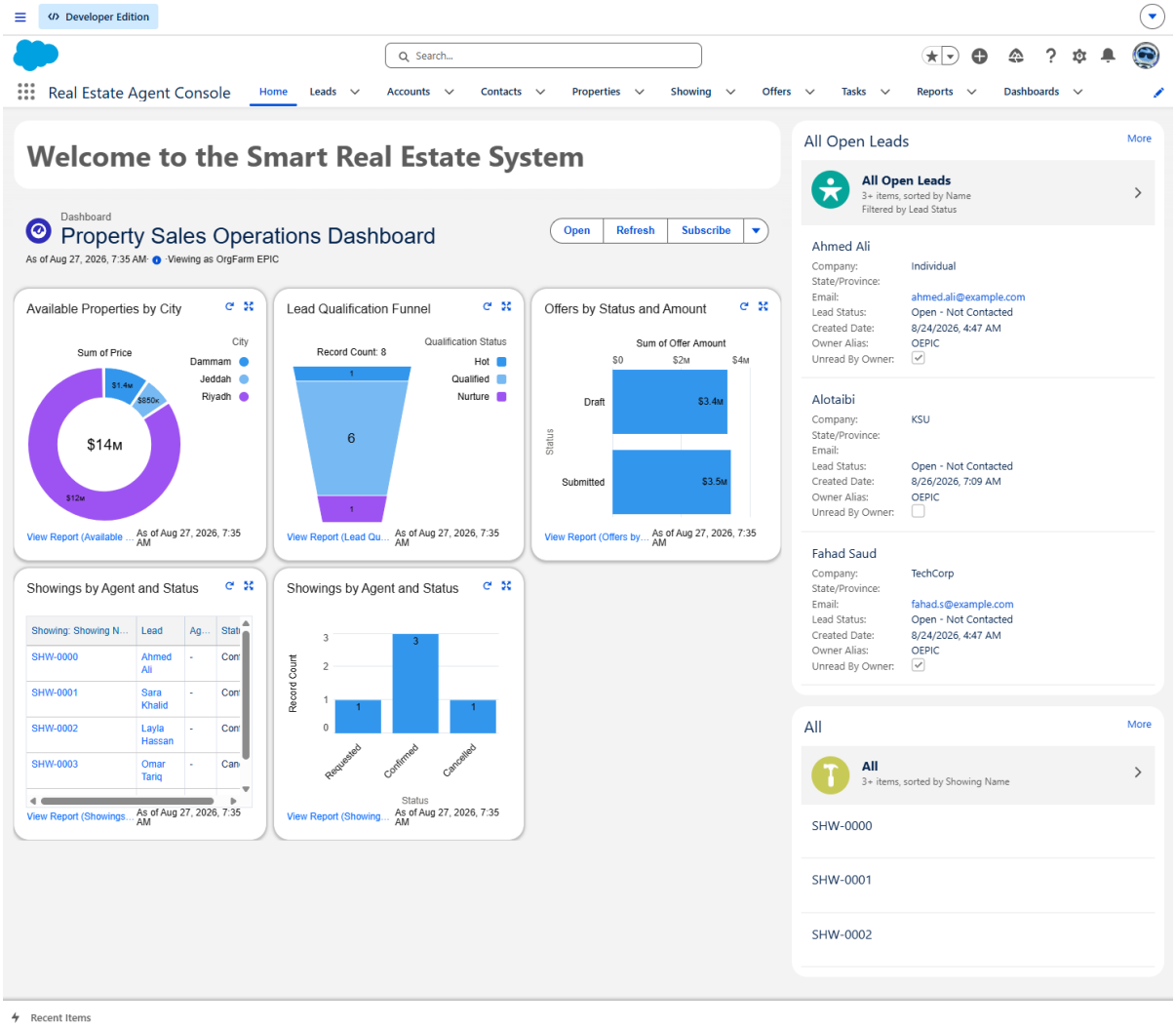


Figure 1. Implemented Real Estate Agent Console and operational dashboard.

2. Scope, Stakeholders, and Requirements

2.1 Scope of Work

In scope: Salesforce Lightning application configuration; lead, property, showing, offer, and task management; Flow-based automation; Agentforce conversational routing and governed actions; a custom LWC; reports and dashboards; and role-based security design.

Out of scope: External customer website integration, payment processing, financial billing or invoicing, post-sale property maintenance, production telephony integration, and third-party property listing feeds.

2.2 Stakeholders and User Personas

Persona	Primary need	Representative user story
Sales Agent	Fast access to qualified leads, available properties, and showing coordination	As a sales agent, I want guided property and scheduling assistance so I can focus on tours and closing activities.
Sales Manager	Inventory, pipeline, offer, and agent-performance visibility	As a sales manager, I need a centralized dashboard so I can monitor operations and identify follow-up priorities.
System Administrator	Governed automation, security, and maintainability	As an administrator, I need controlled configurations and traceable automation so the system remains secure and supportable.

2.3 Functional Requirements

ID	Requirement
FR-01	Maintain property inventory with type, status, price, city, address, bedrooms, bathrooms, area, availability date, and listing agent.
FR-02	Capture lead budget, property preferences, purchase timeline, qualification score, and qualification status.
FR-03	Search or retrieve property details using filters supplied by the user and return only Salesforce records.
FR-04	Update lead preferences through a governed Agentforce action and qualification automation.
FR-05	Check agent availability before a showing is created when an assigned agent is required.
FR-06	Create a Showing record only after property, client, start time, and end time are confirmed.
FR-07	Create or display follow-up tasks for qualified property leads.
FR-08	Route ambiguous requests for clarification, unrelated requests to Off Topic, and human-transfer requests to Escalation.
FR-09	Display operational reports and dashboards for available properties, lead qualification, showings, and offers.
FR-10	Display property information through a custom LWC on a validated Lightning record context.

2.4 Non-Functional Requirements

- Security: users receive only role-appropriate object and field access; negative access tests must be documented.
- Data integrity: required fields, validation rules, and Flow decisions prevent incomplete or invalid transactions.

- Grounded AI behavior: Agentforce must present only records returned by Salesforce actions and must not invent property details.
- Usability: core workflows must be accessible from Lightning pages with clear statuses and actionable feedback.
- Maintainability: Flows, Agentforce instructions, metadata, and LWC source are versioned and documented.
- Auditability: record ownership, created/modified details, task outcomes, and test evidence remain traceable.

3. Solution Architecture

Salesforce Lightning Experience is the primary interface. The Real Estate Agent Console exposes core records, activities, reports, dashboards, and Lightning components. The Property Sales Assistant uses an Agent Router to identify intent and select one of seven specialized subagents. Functional subagents invoke predefined actions backed by Salesforce automation, while governance subagents clarify ambiguous input, reject unrelated requests, or escalate to a person.

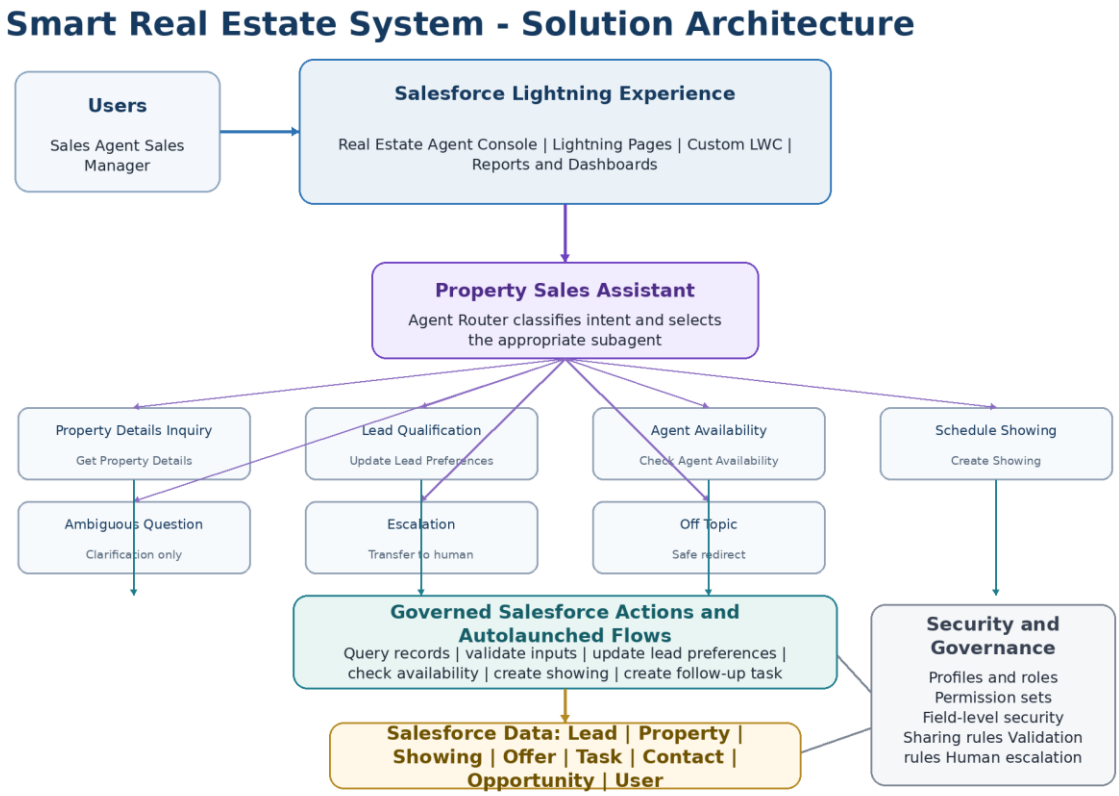


Figure 2. Target solution architecture and governed request flow.

3.1 Architecture Components

Layer	Components	Responsibility
Experience	Real Estate Agent Console, Lightning pages, custom LWC, reports, dashboards	Provide user interaction, record views, operational worklists, and management visibility.
AI orchestration	Property Sales Assistant and Agent Router	Interpret user intent and route the request to the appropriate specialized subagent.
Domain subagents	Property Details, Lead Qualification, Agent Availability, Schedule Showing	Collect required inputs and invoke controlled Salesforce actions.
Governance subagents	Ambiguous Question, Escalation, Off Topic	Request clarification, transfer to a person, or safely redirect unrelated requests.
Automation	Autolaunched Flows, validation logic, task creation	Query or update records, enforce prerequisites, and return structured outcomes.
Data	Lead, Property, Showing, Offer, Task, Contact, Opportunity, User	Store operational records, relationships, ownership, and reporting data.
Security	Profiles, roles, permission sets, FLS, sharing, validation	Restrict access and protect data quality; final evidence requires access testing.

3.2 Agentforce Routing and Actions

Subagent	Purpose	Action or outcome
Property Details Inquiry	Search and present grounded property information	Get Property Details
Lead Qualification	Collect preferences and initiate qualification automation	Update Lead Preferences
Agent Availability	Check an agent for the requested period	Check Agent Availability
Schedule Showing	Create a confirmed property showing	Create Showing after confirmation and availability checks
Ambiguous Question	Ask the user for a clearer real-estate request	No data action
Escalation	Transfer the conversation when human assistance is requested	Escalate to human
Off Topic	Redirect requests unrelated to real estate	No data action

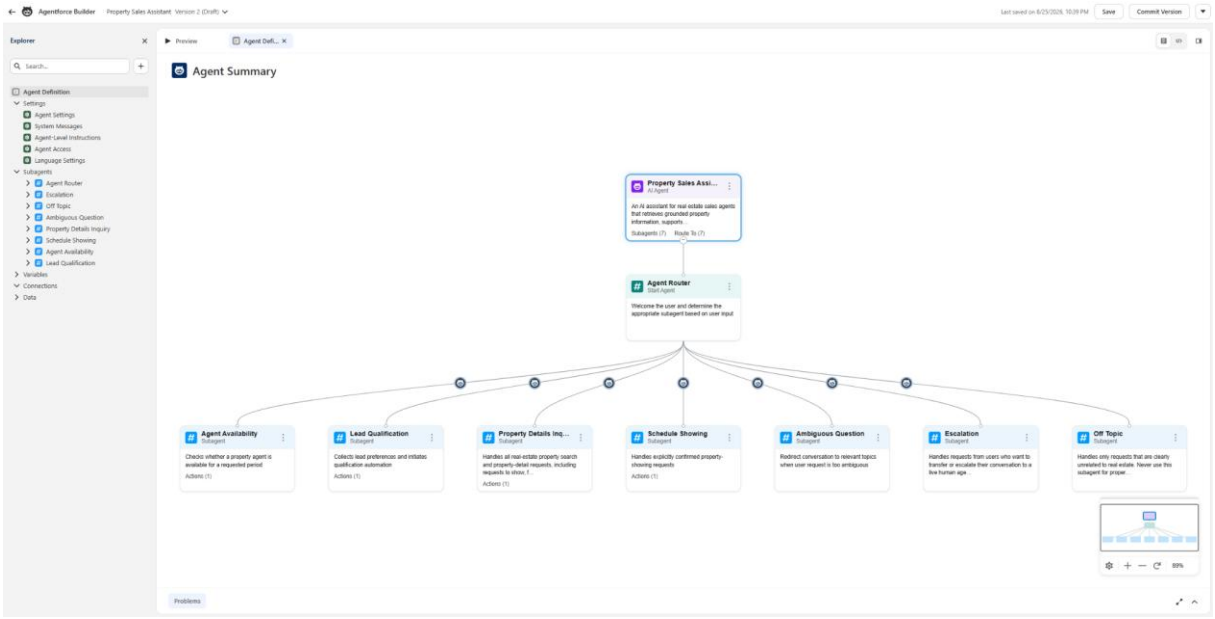


Figure 3. Agentforce Builder routing structure. The supplied evidence shows Version 2 in Draft status.

3.3 Representative End-to-End Flow

Step	System behavior
1	A sales user asks for properties using criteria such as city, budget, bedrooms, or property type.
2	The Agent Router selects Property Details Inquiry, which calls Get Property Details.
3	The action returns matching Property records; Agentforce displays only returned values.
4	If the user selects an available property and confirms a showing, the system collects client and time information.
5	Agent Availability checks the requested period when an agent is required.
6	Schedule Showing calls Create Showing after confirmation; the Flow validates Lead and Property references and creates the record.
7	The new record is available in Lightning pages and downstream reports or dashboards.

4. Data Architecture

The logical model combines Salesforce standard objects with custom real-estate objects. Lead stores qualification and preference data. Property represents inventory. Showing connects a Lead or Contact with a selected Property and assigned User. Offer relates commercial activity to a Property and, where applicable, Contact and Opportunity records. Task supports follow-up work and ownership.

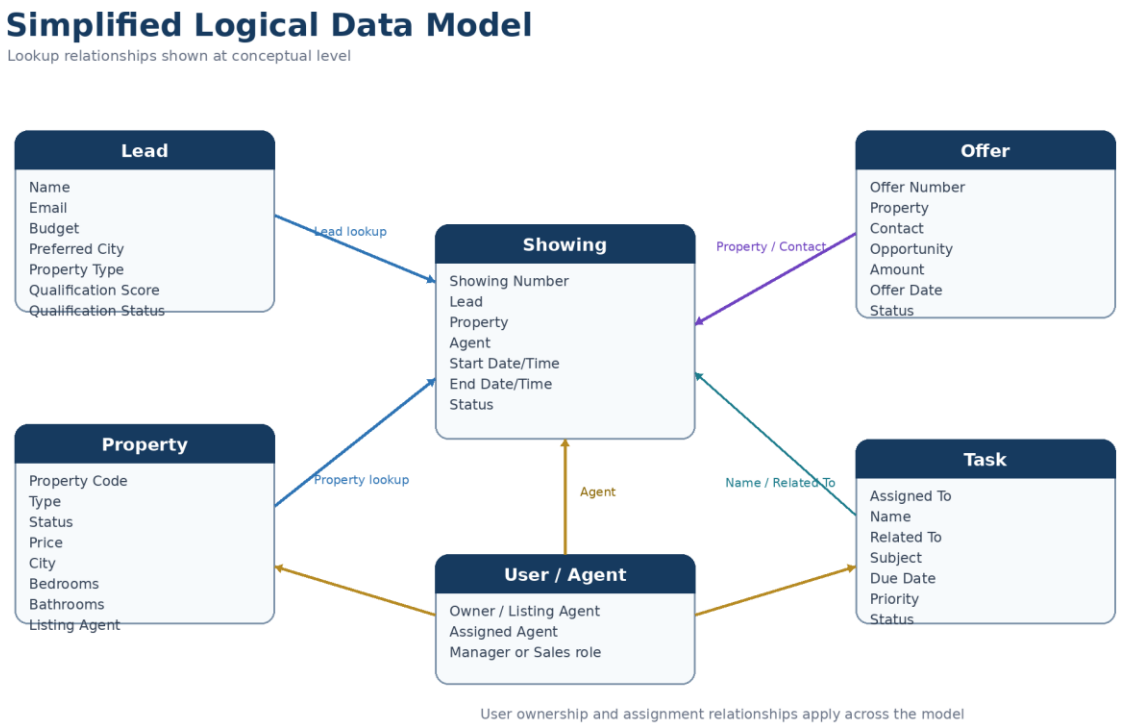


Figure 4. Simplified logical data model for documentation and review.

4.1 Core Object Responsibilities

Object	Purpose	Important relationships
Lead	Capture client identity, budget, preferences, timeline, and qualification results	Referenced by Showing and follow-up Task records
Property	Maintain real-estate inventory and availability details	Referenced by Showing and Offer; owned or listed by User
Showing	Record a scheduled property viewing	Looks up to Property, Lead or Contact, and Agent/User
Offer	Record a proposed amount and status	Looks up to Property and may reference Contact and Opportunity
Task	Track follow-up and operational work	Assigned to User; related to Lead, Contact, Property, or other supported record
User	Represent owners, listing agents, assigned agents, and managers	Ownership and assignment relationships across operational records

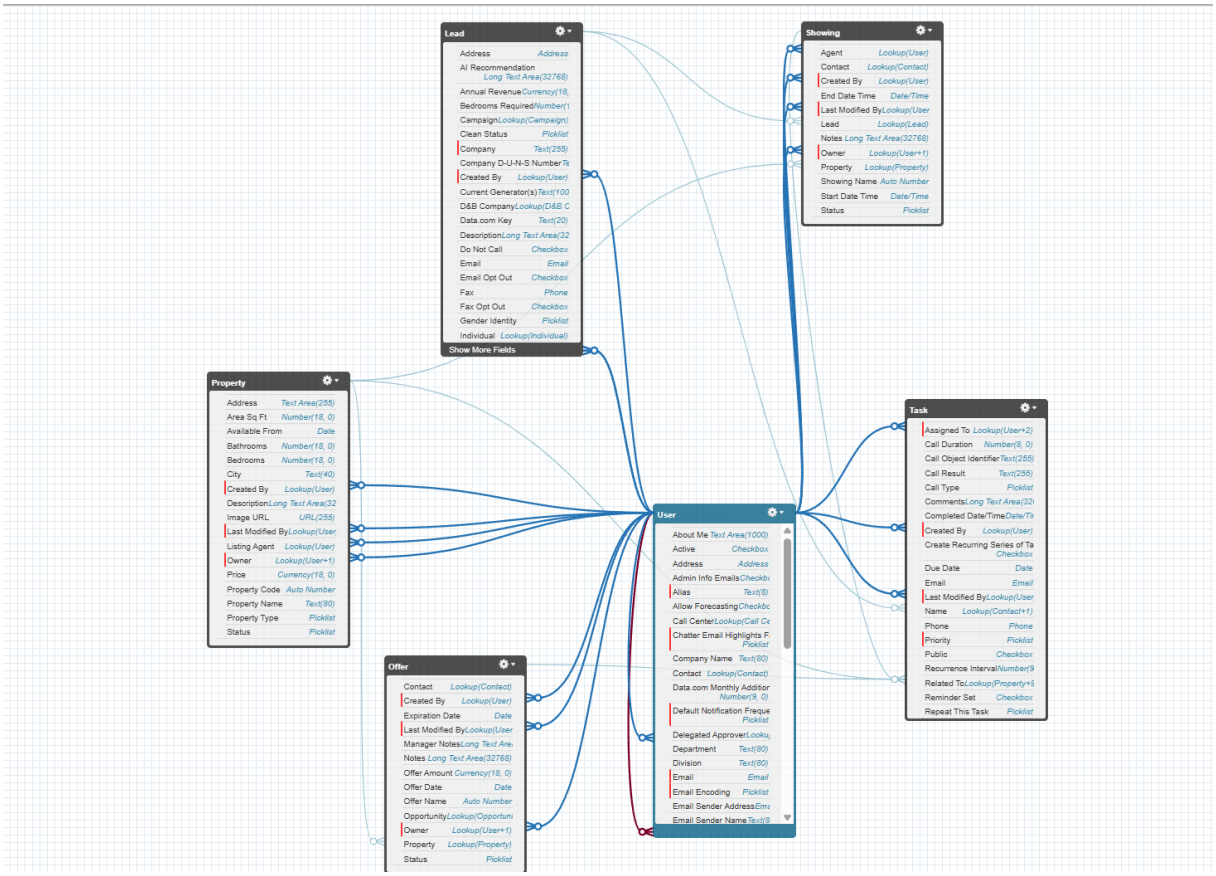


Figure 5. Salesforce Schema Builder implementation evidence.

5. Security and Governance Architecture

The security model is based on Salesforce role-based access controls. Profiles establish baseline permissions, permission sets provide additional access, the role hierarchy and sharing configuration control record visibility, and Field-Level Security restricts sensitive fields. The final submission should include screenshots and negative tests proving the configured access model rather than relying only on design statements.

Control	Design intent	Required evidence
Profiles	Define baseline object and application access for sales and administration users	Profile object-permission screenshot
Permission sets	Grant narrowly scoped capabilities without broad profile changes	Permission set assignment and permissions
Role hierarchy and sharing	Provide managers broader visibility while limiting peer or unrelated records as required	Sharing settings and two-user access test
Field-Level Security	Hide sensitive internal fields from standard sales users	Visible/hidden field comparison
Validation rules	Prevent incomplete, inconsistent, or unauthorized transactions	Negative input and error-message screenshots

Control	Design intent	Required evidence
Agentforce grounding	Use record-returning actions and instructions that prohibit invented property details	Successful, no-result, ambiguous, and off-topic tests
Human escalation	Transfer requests outside the assistant's permitted authority	Escalation test and fallback behavior

5.1 AI Safety Controls

- Property responses must be grounded in records returned by Get Property Details.
- The assistant must not invent properties, prices, addresses, availability, or features.
- Showing creation requires a specific property, client identity, start time, end time, and confirmation.
- Ambiguous requests are clarified rather than answered with assumptions.
- Clearly unrelated requests are redirected without invoking a data action.
- Explicit requests for a person are routed through Escalation, with a defined failure fallback.

6. Project Plan and Delivery Schedule

The following eight-week schedule provides a structured project plan. It documents the intended sequence of work and should not be described as the exact elapsed implementation duration unless it matches the actual course timeline.

Week	Work package	Key outputs
1	Requirements and planning	Problem statement, scope, objectives, personas, user stories, acceptance approach
2	Data and security design	Object model, relationships, required fields, RBAC and FLS design
3	Salesforce configuration	Custom objects and fields, Lightning app, pages, layouts, sample data
4	Automation	Flows, validation rules, qualification logic, task automation, showing validation
5	UI and analytics	Custom LWC, reports, dashboards, responsive Lightning experience
6	Agentforce	Agent Router, seven specialized subagents, actions, grounding and escalation instructions
7	Integration testing and UAT	Positive and negative test cases, security tests, issue resolution
8	Deployment and documentation	Deployment evidence, final documentation, demo script, release checklist

6.1 Deliverables

- Configured Salesforce Real Estate Agent Console with operational records and Lightning pages.
- Agentforce Property Sales Assistant with routed subagents and governed actions.

- Flow and validation automation for lead qualification, availability, scheduling, and follow-up.
- Custom propertyCard LWC source and deployment evidence.
- Reports and dashboard for inventory, leads, showings, and offers.
- Project documentation, test evidence, architecture diagrams, and 5-10 minute demo video.

6.2 Release Gates

Gate	Exit condition	Current evidence position
Configuration complete	Objects, fields, pages, automations, reports, and sample records are present	Representative screenshots supplied; full metadata export is incomplete
Agentforce ready	Version is committed or activated and core routing tests pass	Builder structure supplied; screenshot shows Version 2 Draft
Security verified	Positive and negative user-access tests pass	Design documented; screenshots and tests still required
LWC verified	Component displays correct property fields in its supported record context	Bundle deployed; record-context verification required
Documentation complete	Architecture, test evidence, deployment, and demo materials are reviewed	This document establishes the plan and architecture baseline

7. Risk Management

Risk	Likelihood	Impact	Mitigation and verification
Unsupported or invented AI property details	Medium	High	Ground responses in record-returning actions; prohibit invention; test valid, no-result, ambiguous, and adversarial prompts.
Incomplete or conflicting showing	Medium	High	Require a specific property and client, validate time inputs, check availability, and return exact failure reasons.
Unauthorized record or field access	Low-Medium	High	Apply profiles, permission sets, sharing, and FLS; execute negative tests with sales and manager users.
Poor or duplicate data	Medium	Medium	Use required fields, validation rules, duplicate or matching rules where configured, and controlled data loading.
Agentforce is unavailable or not activated	Low-Medium	Medium	Keep Lightning record pages and manual processes available; commit and activate a tested version before release.
LWC receives the wrong record type	Medium	Medium	Place propertyCard on a Property record page or explicitly resolve the related Property ID when used on another object.
Dashboard becomes stale	Low	Medium	Validate report filters, ownership, and refresh behavior using known sample records before the demo.

8. Test and Acceptance Strategy

Acceptance testing should combine functional, negative, integration, security, and user-interface tests. Each test record should include the precondition, input, expected result, actual result, pass/fail status, and input/output screenshots.

ID	Capability	Acceptance criterion
AC-01	Property search	Only matching Salesforce records are returned; no property is invented.
AC-02	Lead qualification	Preferences are stored and the qualification result is calculated by automation.
AC-03	Qualified-lead follow-up	A correctly assigned follow-up task is created or displayed for a qualified lead.
AC-04	Agent availability	Availability is based on the Flow result for the requested time window.
AC-05	Showing creation	A Showing is created only after required references and times are validated; the Showing number is returned.
AC-06	Invalid input	Missing lead, invalid email, unavailable property, or invalid time input is rejected with a useful message.
AC-07	Off-topic and ambiguity	Unrelated requests are redirected and ambiguous requests trigger clarification without a data action.
AC-08	Escalation	A direct human-transfer request invokes escalation or returns the documented fallback.
AC-09	Security	Sales and manager users see only the records and fields allowed by the approved access matrix.
AC-10	Analytics	Dashboard values agree with the underlying report data and filters.
AC-11	Custom LWC	propertyCard shows the correct Property values on its intended Lightning record context.
AC-12	Regression	Core records and automations continue to work after Agentforce or LWC changes.

8.1 Minimum Demo Data

- At least one available and one unavailable Property record with complete attributes.
- At least one qualified and one non-qualified Lead with different budgets and preferences.
- At least one confirmed, requested, and cancelled Showing record.
- At least one Draft and one Submitted Offer record.
- A sales user and manager user for access comparisons.
- Known dashboard totals that can be reconciled with report results.

9. Deployment, Maintenance, and Troubleshooting

9.1 Deployment Strategy

Org configuration is performed through Salesforce Setup and versioned where metadata is available. The custom LWC is maintained in the SFDX project and deployed through Salesforce CLI. Agentforce changes should be saved, committed as a tested version, and activated only after regression testing. Production release should follow a validate-first deployment, smoke test, and rollback-ready change record.

9.2 Maintenance Approach

- Review Flow errors, failed interviews, and Agentforce action outcomes after changes.
- Reconcile dashboard totals with source reports and verify filters after schema updates.
- Review profiles, permission sets, sharing, and FLS whenever new fields or objects are added.
- Maintain versioned Agentforce instructions and retest grounding, ambiguity, Off Topic, and escalation behavior.
- Run LWC unit tests and record-page smoke tests after component changes.
- Keep a manual Lightning workflow available when Agentforce is unavailable.

9.3 Troubleshooting Priorities

Symptom	Primary checks	Expected corrective action
Agentforce selects the wrong topic	Router and subagent classification instructions; ambiguous examples	Clarify topic scope and add regression prompts
Action returns no record	Action inputs, record IDs/codes, Flow Get Records filters	Correct input mapping or provide a clear no-result response
Showing is not created	Lead email, Property status, required times, availability result, Flow path	Return the exact failure condition and correct invalid data
LWC shows blank values	Record page target, current record type, recordId, field permissions	Use the Property record context or resolve the related Property ID
Dashboard value is incorrect	Report filters, ownership, sample records, refresh timestamp	Correct filters or data and refresh the dashboard
User cannot see a required field	Profile, permission set, FLS, page layout, record sharing	Adjust the least-privilege control and repeat access tests

10. Conclusion

The Smart Real Estate System presents a coherent Salesforce CRM architecture for managing property inventory, lead qualification, showings, offers, tasks, and management reporting. Its strongest differentiator is the governed use of Agentforce: the assistant routes requests to specialized subagents and invokes Salesforce actions instead of relying on unsupported free-form responses. The solution is

suitable for final course documentation once three release items are evidenced: a committed or activated Agentforce version, completed security tests, and a verified propertyCard record context.

> **Final readiness statement:** The architecture and project plan are complete. Final submission readiness depends on attaching test evidence for every major feature and resolving or documenting the identified release gates.

Appendix A. Evidence Inventory

Evidence	What it supports	Documentation use
Real Estate Agent Console dashboard	Implemented Lightning experience, reports, dashboards, leads, showings, and offers	Figure 1 and demo overview
Agentforce Builder summary	Agent Router and seven specialized subagents	Figure 3 and AI architecture
Subagent instruction screenshots	Actions and governance behavior	Agentforce testing and demo evidence
Salesforce Schema Builder	Object fields and lookup relationships	Figure 5 and data model evidence
SFDX propertyCard bundle	Custom LWC source and deployment records	UI customization and deployment documentation
Lead, Property, Showing, Offer, and Task screenshots	Representative records and relationships	Functional test input/output evidence

Appendix B. Submission Completion Checklist

- Replace or confirm the prepared-by name using the official university spelling.
- Add course, section, instructor, student ID, and institution details if required by the submission template.
- Commit or activate the tested Agentforce version and capture the status screenshot.
- Capture profiles, permission sets, role hierarchy, sharing, and FLS evidence.
- Verify propertyCard on the intended record context and capture a populated component screenshot.
- Prepare input and output screenshots for every acceptance criterion used in the final report.
- Export the final document to PDF and verify headings, captions, tables, images, page numbers, and TOC.
- Record the 5-10 minute demo using the student's own voice at 720p or higher.